

Threat Model: Tall Ships Halifax

title: "Insecure Design: Threat Modeling - Tall Ships Halifax" author: [Bhavik Kantilal Bhagat, Miguel Angel Palafox Gomez, Nikola Kriznar] date: "April 2026" geometry: "left=0.5in,right=0.5in,top=0.5in,bottom=0.5in" output: pdf_document

Part 1: Decomposition

1. Threat Model Information

This report models the Tall Ships Halifax ticketing platform as a stateful maritime event workflow rather than a simple e-commerce checkout. The architecture and threat assumptions are derived from the team decomposition work, with emphasis on strict transition integrity, operational safety, and trust-boundary enforcement in the SMU deployment environment.

Field	Value
Application Name	Tall Ships Halifax - Event Ticketing and Tour Booking Platform
Application Version	1.0 (modeled from OWASP Juice Shop v19.2.1 architecture)
Description	Tall Ships Halifax is a stateful ticketing platform that manages harbor cruises, tall ship tours, and seasonal events. The platform uses an Angular SPA frontend and a Node.js/Express backend with JWT-based authentication and SQLite persistence at /var/lib/tallships/db/ticketing.sqlite.
Document Owners Academic Context	Technical Lead, Security Analyst, System Architect SMU and NSCC threat modeling assignment

2. External Dependencies (SMU and NSCC Context)

The following dependencies define critical external and infrastructural relationships that influence the system attack surface and control design.

ID	Dependency	Description
2.1	Node.js Runtime on SMU Linux Infrastructure	Hosts the API and business logic within the campus-managed environment.
2.2	SQLite Persistence Layer	Central data store for users, tickets, lifecycle states, and capacity counters at /var/lib/tallships/db/ticketing.sqlite.
2.3	External Payment Gateway	Third-party payment processor used for reservation settlement and callback verification.
2.4	SMU SMTP Relay	Outbound mail service for ticket confirmations, refund notifications, and operational communications.
2.5	Campus Security Perimeter	Campus firewall and TLS termination at load balancer protecting ingress traffic.

2.1 Architecture Overview *Supplementary Figure: High-level architecture diagram of the Tall Ships Halifax platform and trust boundaries.*

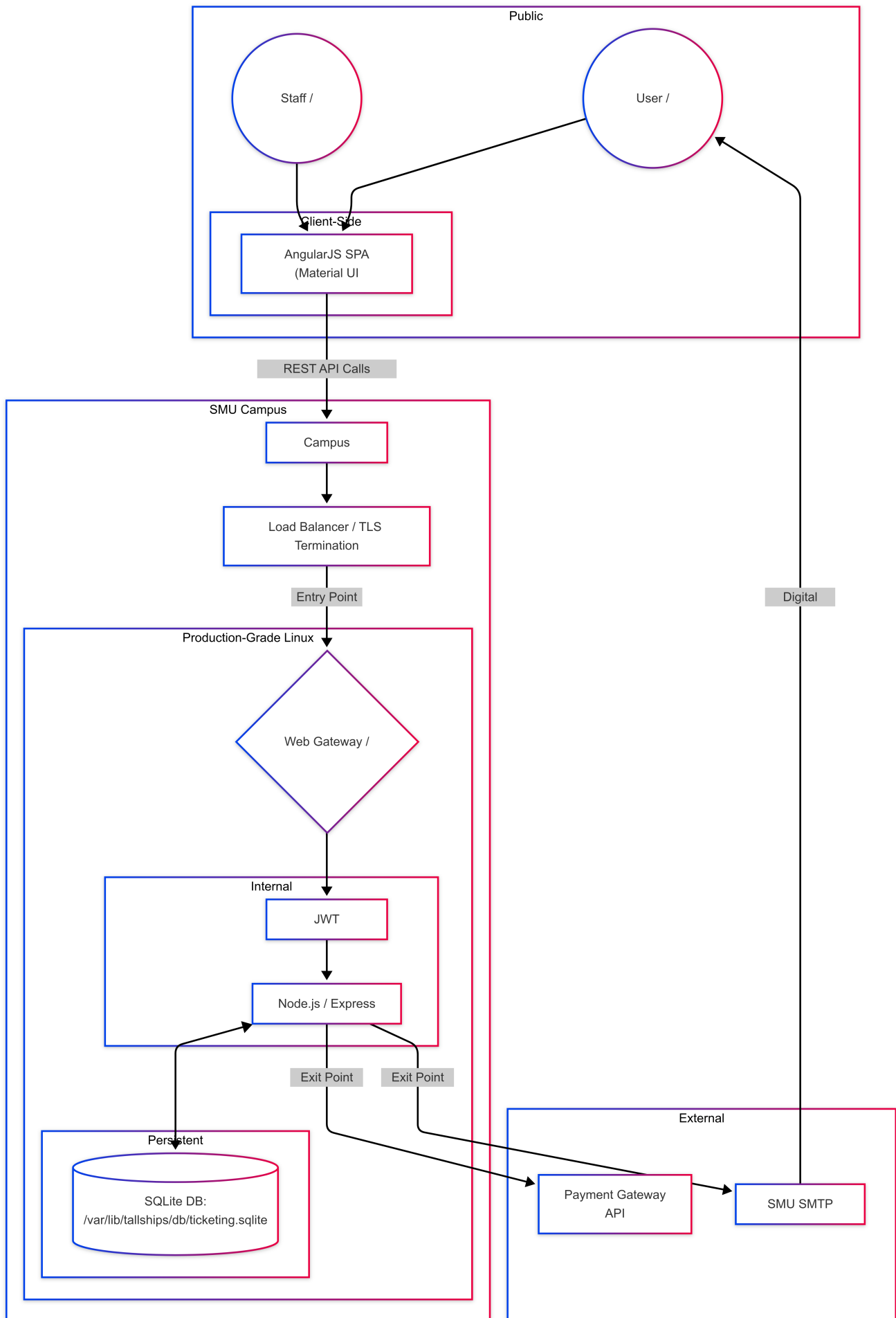


Figure 1: Threat_Modeling_Architecture_Diagram.png

3. Entry and Exit Points

Entry and exit points are the primary locations where data crosses trust boundaries. These interfaces are central to both abuse-case analysis and control placement.

3.1 Entry Points

ID	Name	Description	Trust Levels
3.1.1	Web Gateway (HTTPS)	Primary TLS entry for browser and mobile client traffic.	T1, T2, T5
3.1.2	Event Browsing Endpoint	Public event and schedule browsing APIs.	T1, T2
3.1.3	Reservation API	Moves ticket state from AVAILABLE to RESERVED after server validation.	T2, T5
3.1.4	Payment Endpoint	Processes payment confirmation and transition from RESERVED to PAID.	T2, T5, T7
3.1.5	Refund Request Endpoint	Processes policy-checked transition from PAID to REFUNDED.	T2, T5
3.1.6	Staff Check-in Endpoint	Staff-only endpoint for PAID to USED validation at event access control.	T3, T5

3.2 Exit Points The following technical exits describe where data leaves the SMU-hosted application stack toward external systems or untrusted client environments.

ID	Name	Description	Data Leaving System
3.2.1	SMU SMTP Relay	Outbound ticket and refund communication channel.	Email, ticket reference, event metadata, confirmation timestamp
3.2.2	Payment Request Channel	Outbound API request and callback handling with external payment provider.	Amount, order reference, correlation ID, tokenized payment reference
3.2.3	API Response Channel	Responses to web and mobile clients from Node.js/Express APIs.	Sanitized business payloads, status codes, validation outcomes
3.2.4	Client-side Session Storage	Browser-managed state in Angular SPA.	JWT token, client session metadata, local UI cache

4. Assets

Assets are components of the system that have value and must be protected from unauthorized access, modification, or disruption. They include data, processes, and service capabilities that, if compromised, can affect confidentiality, integrity, availability, financial outcomes, and operational safety.

ID	Asset	Description	Primary Trust Levels	Sensitivity	Business Risk
A1	Ticket Records	Digital ticket entities tied to users and events	T2, T5, T6	High	Fraud and unauthorized entry if modified
A2	Ticket State Integrity	Enforced lifecycle: AVAILABLE to RESERVED to PAID to USED, plus PAID to REFUNDED	T3, T5, T6	Critical	State abuse can bypass payment and enable ticket replay
A3	Payment Information	Transaction metadata and settlement confirmation state	T5, T6, T7	High	Financial loss and legal exposure
A4	User Accounts and Credentials	Authentication identity data and account context	T2, T5, T6	High	Account takeover and workflow abuse
A5	JWT Authentication Tokens	Access tokens for user and staff operations	T2, T3, T8	High	Identity spoofing and privilege abuse
A6	Promotion and Pricing Rules	Discount eligibility and policy logic	T2, T5	Medium-High	Revenue loss from unauthorized discounts
A7	Event Capacity Data	Remaining seats and booking counters per event	T3, T5, T6	High	Overselling and safety/compliance impact
A8	Audit and Transaction Logs	Traceability for booking, refund, and check-in operations	T4, T5, T6	High	Repudiation and incident response gaps
A9	Staff Check-in Operations	Operational controls for event gate validation	T3, T5	High	Invalid ticket acceptance or denial of valid entry
A10	Application Database	Persistent platform data store	T5, T6	Critical	Broad confidentiality and integrity compromise
A11	Backend APIs	Service layer implementing business rules and state controls	T5	High	Central attack surface for abuse
A12	Hosting and Container Environment	Runtime infrastructure and service configuration	T4, T5, T10	Medium-High	Lateral movement and configuration abuse

5. Trust Levels

Trust levels represent users and system components that interact with the platform and the degree of confidence assigned to each. They provide the basis for access design, control depth, and threat impact analysis across the full ticket lifecycle.

ID	Entity	Description	Trust Level	Capabilities	Risk if Compromised
T1	Guest User	Unauthenticated user browsing events	Low	Read-only browsing and catalog queries	Reconnaissance and abuse probing
T2	Registered Customer	Authenticated customer account	Medium	Reservations, payments, refunds, ticket management	Workflow and financial abuse
T3	Staff User	Operational event staff	High	Check-in validation and attendance management	Unauthorized overrides and access abuse
T4	Administrator	Platform administrator role	Very High	User, policy, and platform configuration control	Full system compromise
T5	Backend Application	Node.js/Express business logic layer	High	Authn/authz checks, state transitions, external integration	End-to-end integrity failure
T6	Database System	SQLite persistence layer	High	Persistent storage for all core entities	Mass tampering or data leakage
T7	External Payment Service	Third-party payment infrastructure	Medium-High	Payment authorization and callback signaling	Forged payment completion
T8	JWT Authentication Mechanism	Token issuance and validation controls	High	Session and role trust assertions	Token forgery and impersonation
T9	Angular Client Application	Frontend execution environment	Low (Untrusted)	UX rendering and API invocation	Client-side manipulation
T10	Hosting Environment	Campus-hosted runtime and operations platform	Medium	Service execution, deployment, networking	Privilege escalation and service disruption

6. Level 1 DFD and Decomposition

6.1 Level 1 DFD Narrative (10-Step Technical Flow) The Level 1 decomposition models ticketing as a stateful workflow in which untrusted requests are transformed into validated state changes inside the SMU trust boundary before any external propagation occurs.

1. User and staff traffic enters through the HTTPS web gateway within the SMU perimeter.
2. Requests route to Node.js/Express API handlers for browse, reserve, pay, refund, and check-in functions.
3. JWT controls verify signature, expiration, audience, and role claims before business logic execution.
4. Endpoint authorization enforces role constraints, especially for staff-only workflows.
5. Business pre-validation checks event identity, ticket ownership, legal transition, payload integrity, and policy eligibility.
6. The API reads current persisted state and capacity from `/var/lib/tallships/db/ticketing.sqlite`.
7. Transition logic applies an atomic state change only when all preconditions are satisfied.

8. Payment flows invoke outbound provider calls and accept callbacks only after cryptographic and semantic verification.
9. Successful PAID and REFUNDED transitions trigger notifications through the SMU SMTP relay.
10. The platform returns sanitized responses and writes immutable audit events to support non-repudiation.

6.1.1 SMU Trust Boundary Enforcement The SMU trust boundary separates untrusted actors (public clients and external provider endpoints) from trusted execution and persistence components (Express services, SQLite storage, and operational logging).

- Boundary ingress controls: TLS, route exposure constraints, request-size controls, and rate limiting.
- Boundary internal controls: server-side state machine enforcement, role checks, and transaction integrity validation before commit.
- Boundary egress controls: data minimization for SMTP, payment, and API responses; exclusion of internal diagnostics from client-visible payloads.

This separation prevents client-authoritative state mutation by ensuring only trusted server code can authorize and persist lifecycle transitions.

6.2 Finite State Machine Logic for Lifecycle Integrity The ticket lifecycle is enforced as a constrained state machine:

AVAILABLE -> RESERVED -> PAID -> USED
PAID -> REFUNDED (controlled reversal)

Required controls include: - Server-authoritative transition validation against current persisted state. - Atomic compare-and-set updates to prevent race-condition double booking or double check-in. - Idempotency and replay protection for payment callbacks and refund operations. - Role enforcement for staff check-in actions. - Immutable audit telemetry for accountability and dispute resolution.

Insecure design failure modes to avoid include client-authoritative transition requests, non-atomic state updates, missing callback/refund idempotency controls, shallow staff authorization, and audit gaps that prevent dispute reconstruction.

Technical position: primary risk concentration in this system is business logic integrity. Every transition must be validated server-side against persisted current state, policy constraints, and actor authorization before commit.

Figure 1: Level 1 DFD of the Tall Ships Halifax Ticketing System.

Part 2: Determine and Rank

7. STRIDE Analysis

STRIDE is applied to classify security threats by attack intent and impact domain. The framework supports structured identification of identity abuse, unauthorized modification, traceability failures, data exposure, service disruption, and privilege escalation.

- Spoofing: impersonation through stolen credentials or tokens.
- Tampering: unauthorized modification of data, request parameters, or lifecycle state.
- Repudiation: denial of actions due to insufficient or unreliable auditability.
- Information Disclosure: exposure of sensitive information to unauthorized entities.
- Denial of Service: degradation or interruption of service availability.
- Elevation of Privilege: unauthorized gain of higher access rights.

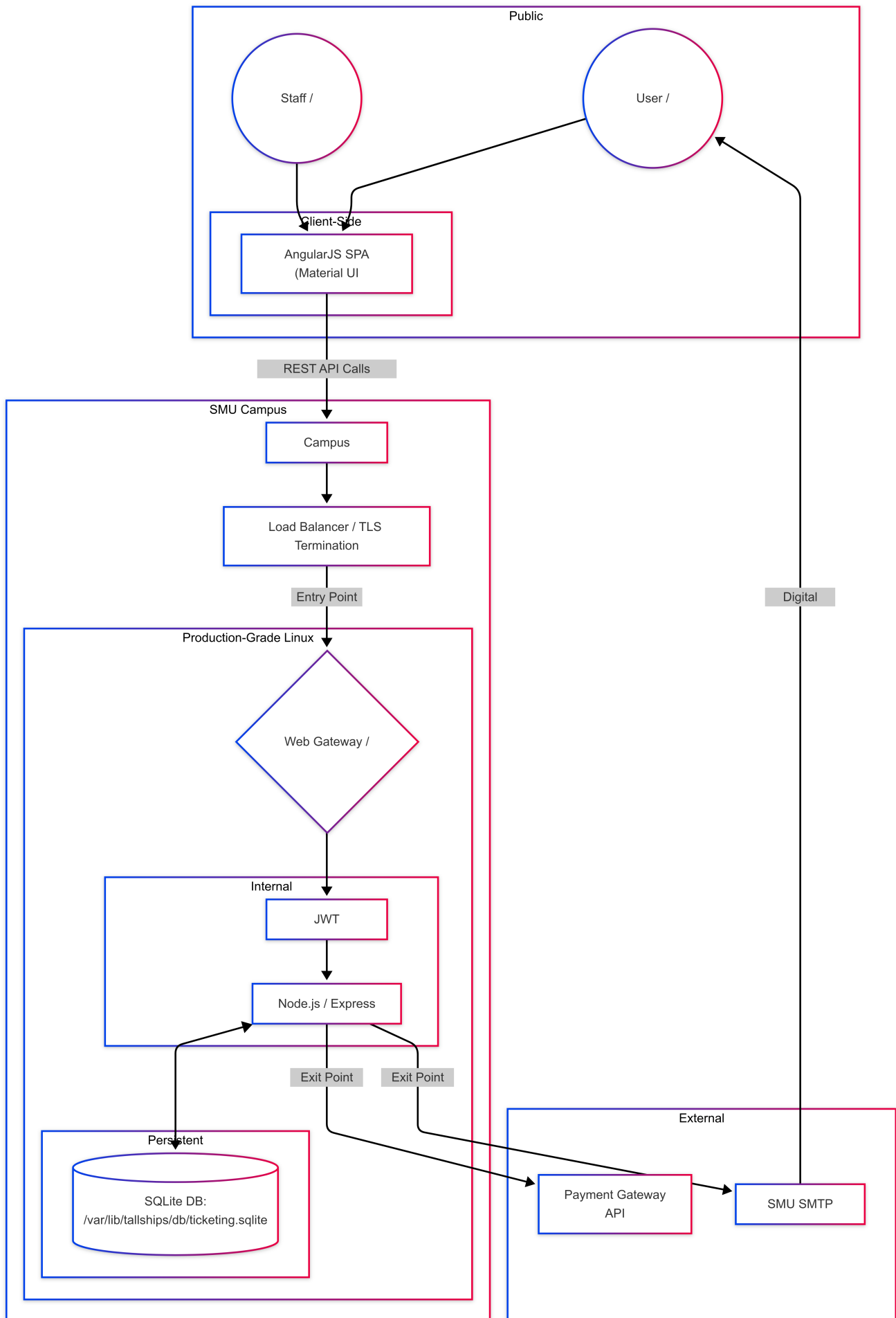


Figure 2: Threat_Modeling_DataFlow_Diagram.png

ID	Threat Type	Threat Description	Security Controls	Asset Affected	Trust Levels	Impact
S1	Spoofing	Attacker steals or forges JWT to impersonate a valid user or staff member	Short-lived tokens, secure storage, strong signing key management, mandatory token validation	A5	T2, T3, T8	Unauthorized access and privilege abuse
S2	Spoofing	Attacker logs in with stolen credentials	Strong password controls, MFA, rate limiting, lockout policy	A4	T2	Unauthorized purchases and refunds
T1	Tampering	User manipulates requests to force illegal ticket transition	Server-side state machine enforcement and strict transition validation	A2	T2, T5	Payment bypass and lifecycle fraud
T2	Tampering	User applies unauthorized discount logic	Server-side promo validation and policy-bound redemption controls	A6	T2, T5	Revenue loss
T3	Tampering	Capacity counters are altered to oversell events	Atomic transactions, integrity constraints, server-side capacity guards	A7	T3, T5, T6	Safety and operational disruption
T4	Tampering	Payment response is forged to simulate successful payment	Callback signature validation and trusted provider verification	A3	T5, T7	Fraudulent ticket issuance
R1	Repudiation	User denies an operation due to weak traceability	Tamper-resistant logging with actor identity and timestamps	A8	T2, T5	Dispute and compliance failures
R2	Repudiation	Staff denies check-in action	Staff identity-bound operational audit trails	A9	T3	Accountability failure

ID	Threat Type	Threat Description	Security Controls	Asset Affected	Trust Levels	Impact
I1	Information Disclosure	User account data is exposed	Access controls, encrypted transport, data minimization	A4	T5, T6	Privacy breach
I2	Information Disclosure	Payment-related data is leaked	Provider tokenization, minimal storage, encryption controls	A3	T5, T6, T7	Financial and legal impact
I3	Information Disclosure	Sensitive values leak through logs	Structured log hygiene and restricted log access	A8	T5, T6	Internal data exposure
D1	Denial of Service	API endpoints are flooded	Rate limiting, throttling, and load balancing	A11	T1, T2	Service unavailability
D2	Denial of Service	Reservation abuse blocks inventory	Reservation expiry controls and reservation count limits	A7	T2	Booking denial for legitimate users
E1	Elevation of Privilege	User escalates to staff operations	Strict RBAC and endpoint authorization checks	A9	T2, T5	Unauthorized operational actions
E2	Elevation of Privilege	Backend flaws allow admin privilege gain	Secure coding practices, strict validation, continuous testing	A11	T2, T5	Full system compromise

Figure 2: Finite State Machine for Ticket Lifecycle Validation.

8. OWASP Risk Rating Model

The OWASP Risk Rating approach was used to prioritize STRIDE threats through two scored dimensions: Severity and Likelihood. Each dimension uses five yes/no criteria, where Yes contributes 1 point and No contributes 0 points.

Severity represents business and operational impact, including financial loss, reputational harm, ticket state integrity failure, workflow disruption, and scale of affected users.

Likelihood represents exploit feasibility, including remote exploitability, required privileges, automation potential, attacker skill threshold, and tooling complexity.

Final Risk Score = Severity + Likelihood (range 0-10).

8.1 Severity Questions

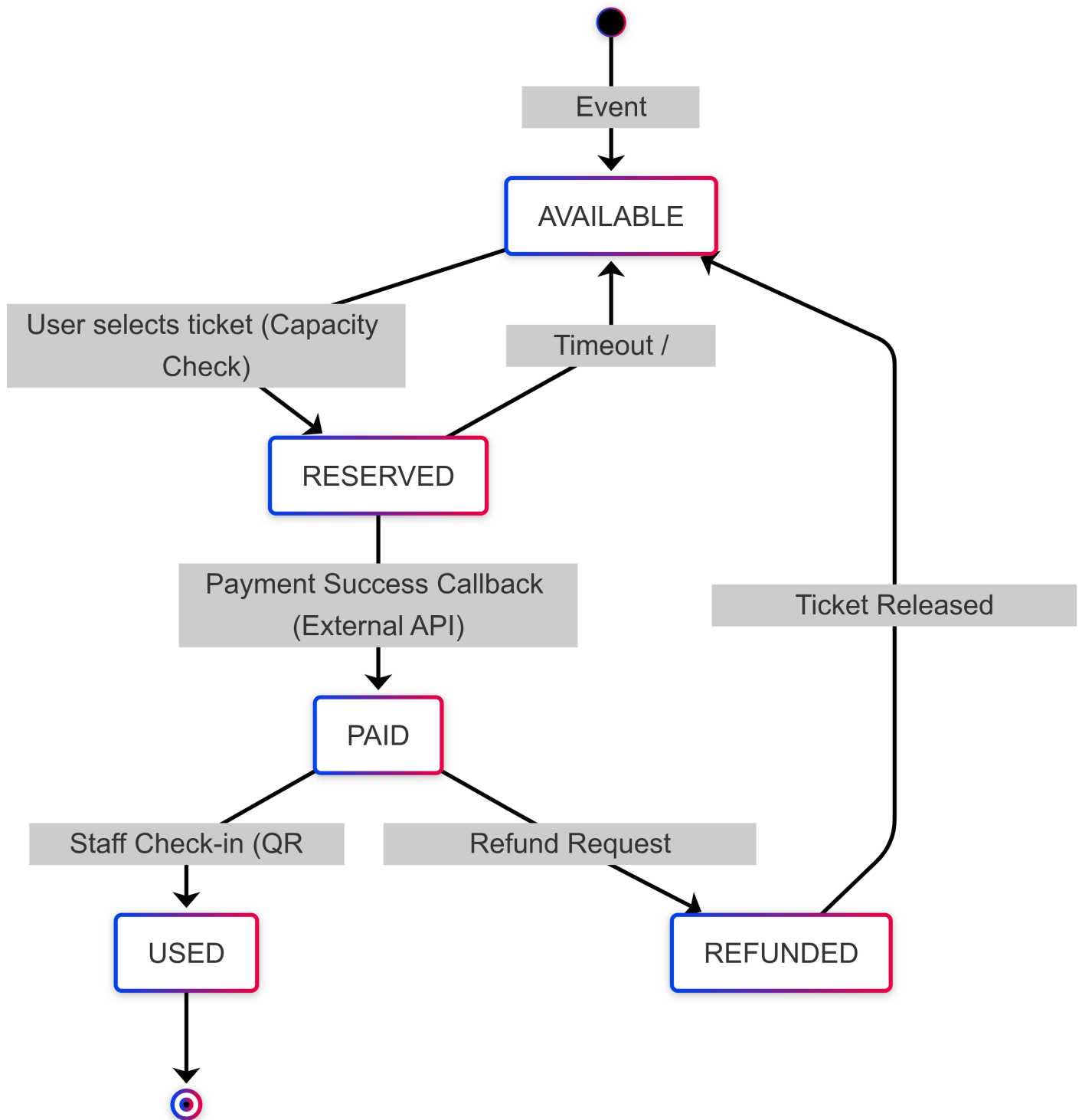


Figure 3: Threat_Modeling_State_Diagram.png

ID	Question
Z1	Does the threat lead to direct financial loss (for example, free tickets or refund abuse)?
Z2	Does the threat cause reputational damage (for example, overselling events or customer complaints)?
Z3	Does the threat affect ticket state integrity (for example, payment bypass or ticket replay)?
Z4	Does the threat disrupt core workflows (booking, refund, check-in)?
Z5	Does the threat impact multiple users or events at scale?

8.2 Likelihood Questions

ID	Question
L1	Can the threat be exploited remotely over the internet?
L2	Can it be exploited without authentication or with low privileges?
L3	Can the attack be automated or repeated easily?
L4	Does it require low technical skill?
L5	Can simple tools perform the exploit?

8.3 STRIDE Risk Scoring Table

ID	Severity (Z1-Z5)	Likelihood (L1-L5)	Risk Score
S1	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✗ L3✓ L4✓ L5✓ = 4	8
S2	Z1✓ Z2✓ Z3✗ Z4✓ Z5✗ = 3	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	8
T1	Z1✓ Z2✓ Z3✓ Z4✓ Z5✓ = 5	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	10
T2	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	9
T3	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✗ L2✗ L3✗ L4✓ L5✓ = 2	6
T4	Z1✓ Z2✓ Z3✓ Z4✓ Z5✓ = 5	L1✓ L2✗ L3✓ L4✓ L5✓ = 4	9
R1	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	9
R2	Z1✗ Z2✓ Z3✗ Z4✓ Z5✗ = 2	L1✗ L2✗ L3✗ L4✓ L5✓ = 2	4
I1	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✗ L3✓ L4✓ L5✓ = 4	8
I2	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✗ L3✓ L4✓ L5✓ = 4	8
I3	Z1✗ Z2✓ Z3✗ Z4✓ Z5✓ = 3	L1✗ L2✗ L3✗ L4✓ L5✓ = 2	5
D1	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	9
D2	Z1✓ Z2✓ Z3✗ Z4✓ Z5✓ = 4	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	9
E1	Z1✓ Z2✓ Z3✓ Z4✓ Z5✓ = 5	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	10
E2	Z1✓ Z2✓ Z3✓ Z4✓ Z5✓ = 5	L1✓ L2✓ L3✓ L4✓ L5✓ = 5	10

8.4 Risk Ranking

Rank	Threat ID	Risk Score
1	T1	10
1	E1	10
1	E2	10
2	T2	9
2	T4	9
2	R1	9
2	D1	9
2	D2	9

Rank	Threat ID	Risk Score
3	S1	8
3	S2	8
3	I1	8
3	I2	8
4	T3	6
5	I3	5
6	R2	4

Part 3: Countermeasures

9. Technical Mitigations for Top-Ranked Threats

The following mitigation plan targets the top-ranked threats (T1, E1, E2, T2, T4) and emphasizes security-by-design controls for lifecycle integrity, capacity safety, payment trust, and staff-operated check-in enforcement.

Threat ID	Design Principle	OWASP Cheat Sheet Reference	Juice Shop CVE Reference	Mitigation Plan
T1	Defense in Depth; Never Trust Client Input	OWASP Input Validation Cheat Sheet; OWASP Transaction Authorization Cheat Sheet	CVE-2020-36604 (prototype pollution risk affecting integrity in Juice Shop dependency stack)	Implement a server-owned finite state transition service. Require current-state predicates in SQL updates, reject illegal transitions, and log all transitions with actor identity and correlation ID.
E1	Least Privilege	OWASP Authorization Cheat Sheet; OWASP REST Security Cheat Sheet	CVE-2020-15084 (JWT-related trust weakness in Juice Shop advisory context)	Enforce deny-by-default RBAC middleware on all staff routes, including PAID to USED check-in endpoints. Resolve effective role server-side for high-risk operations and hard-fail ambiguous role context.
E2	Secure by Default; Fail Securely	OWASP Authentication Cheat Sheet; OWASP Secure Coding Practices Quick Reference Guide	CVE-2020-15084 (JWT algorithm weakness demonstrates authn/authz dependency risk)	Segregate admin route groups, enforce token issuer and audience checks, apply short token lifetime and key rotation, and block deployment on high-severity SAST findings.

Threat ID	Design Principle	OWASP Cheat Sheet Reference	Juice Shop CVE Reference	Mitigation Plan
T2	Complete Mediation	OWASP Business Logic Security Cheat Sheet; OWASP Mass Assignment Cheat Sheet	CVE-2021-21366 (dependency vulnerability pattern relevant to unsafe input handling paths)	Compute all final pricing on the server, bind promo codes to event and policy constraints, and enforce one-time or bounded redemption using database constraints and counters.
T4	Trust but Verify	OWASP Webhook Security Cheat Sheet; OWASP Transaction Authorization Cheat Sheet	CVE-2022-24434 (malformed multipart payload handling risk shown in Juice Shop test coverage)	Validate payment callback signature (HMAC-SHA256), verify nonce and timestamp, enforce idempotency keys, and perform payment verification plus state transition in one atomic transaction.

Implementation note: All state-changing operations must remain server-authoritative and auditable. The SQLite source of truth at `/var/lib/tallships/db/ticketing.sqlite` and outbound SMU SMTP relay notifications are treated as controlled trust-boundary operations.

Part 4: Conclusion and SAST

10. OWASP Top 10 Mapping and SAST Analysis

10.1 Threat Mapping to OWASP Top 10 (2021)

Threat IDs	Primary OWASP Category	Rationale
E1, E2	A01: Broken Access Control	Privilege escalation and unauthorized access to staff/admin operations are direct access control failures.
S1, S2	A07: Identification and Authentication Failures	Token and credential abuse target weak identity assurance and session trust.
T1, T2, T4	A04: Insecure Design	Illegal state transitions, discount tampering, and weak callback trust indicate missing secure workflow design controls.
I1, I2, I3	A02: Cryptographic Failures and A09: Security Logging and Monitoring Failures	Sensitive data exposure and log leakage indicate confidentiality and observability control gaps.
D1, D2	A05: Security Misconfiguration	Missing throttling, reservation guardrails, and operational limits can create availability failures.

Threat IDs	Primary OWASP Category	Rationale
R1, R2	A09: Security Logging and Monitoring Failures	Weak traceability undermines non-repudiation and incident investigation quality.

10.2 SAST Scan Analysis: datacache.ts Static analysis and manual review of datacache.ts indicate a high-risk design pattern where mutable singleton data structures are exported and shared across modules. This creates three major security consequences:

1. Shared mutable global state can be modified from any importing module, causing cross-request integrity side effects.
2. Direct object exposure removes encapsulation and allows unauthorized mutation paths that can bypass intended authorization boundaries.
3. In-memory state dependence introduces restart inconsistency and race-condition risks that can impact reservation and capacity integrity.

These findings map primarily to: - A01: Broken Access Control - A04: Insecure Design - A08: Software and Data Integrity Failures

Recommended remediation trajectory for datacache.ts: - Replace raw exported mutable objects with a controlled repository/service interface. - Enforce authorization-aware mutation methods. - Use immutable read views and defensive cloning. - Reconcile in-memory state with durable persistence and transactional update semantics.

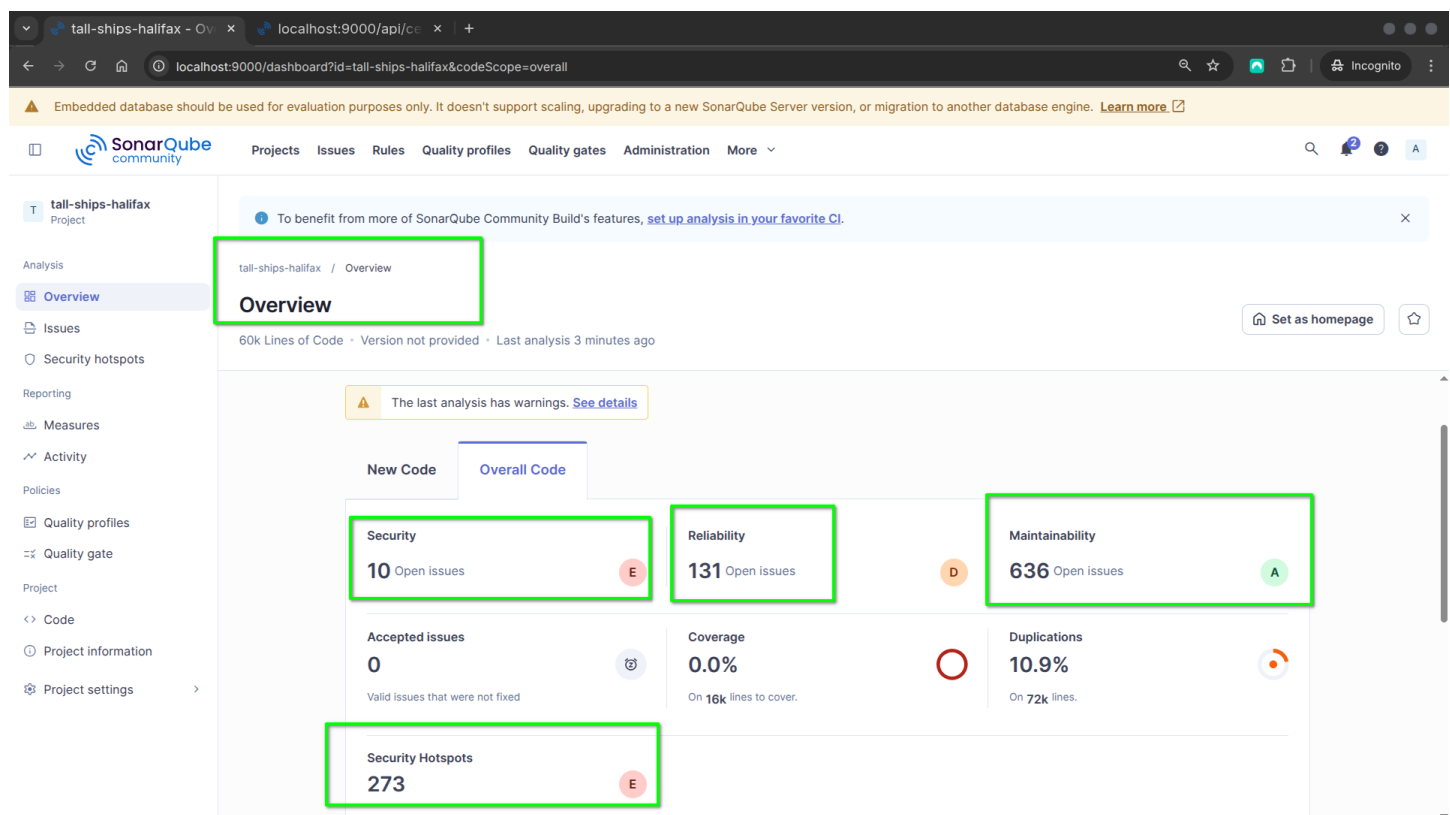


Figure 4: sonarqube_overall_analysis.png

Figure 3: SonarQube Dashboard showing Security “E” Rating.

Figure 4: Detailed SAST Issues in datacache.ts.

10.3 Final Conclusion The threat model shows that Tall Ships Halifax risk is concentrated in business work-flow integrity, authorization depth, and trust-boundary controls rather than only traditional input-level flaws.

Embedded database should be used for evaluation purposes only. It doesn't support scaling, upgrading to a new SonarQube Server version, or migration to another database engine. [Learn more](#)

Security Hotspots

0.0% Security Hotspots Reviewed

To review Acknowledged Fixed Safe

273 Security Hotspots to review

Review priority: High

- Authentication 179
- Cross-Site Scripting (XSS) 9

Review priority: Medium

- Code Injection (RCE) 9
- Denial of Service (DoS) 10
- Permission 2
- Weak Cryptography 11

Review this potentially hard-coded password.

Hard-coded passwords are security-sensitive [typescript:S2068](#)

Status: To review
This security hotspot needs to be reviewed to assess whether the code poses a risk. [Review](#)

Where is the risk? What's the risk? Assess the risk How can I fix it? Activity

frontend/src/app/Services/two-factor-auth-service.spec.ts [Open in IDE](#)

```

55 it('should set up 2FA directly via the rest api', inject([TwoFactorAuthService, HttpTestingController],
56   fakeAsync((service: TwoFactorAuthService, httpMock: HttpTestingController) => {
57     let res: any
58     service.setup('s3cr3t!', 'initialToken', 'setupToken').subscribe((data) => (res = data))
59
60     const req = httpMock.expectOne('http://localhost:3000/rest/2fa/setup')
61     req.flush({})
62     tick()
63
64     expect(req.request.method).toBe('POST')
65     expect(req.request.body).toEqual({ password: 's3cr3t!', initialToken: 'initialToken', setupToken: 'setupToken' })

```

Figure 5: sonarqube_security_hotspots.png

By enforcing strict state-machine validation, robust RBAC, callback authenticity checks, and consistent auditability, the platform can materially reduce the highest-ranked risks while preserving operational reliability for ticketing and event safety.